

1. Scenario and Model Params

- Walkable domain Ω_w and channel geometry
- Obstacles and internal entrances
- Behavioral preference $\hat{\mathbf{p}}$
- Geometric guidance parameter η_0

2. Control Variables

Direction configuration $\mathbf{s}$
+1: forward one-way
-1: reverse one-way
0: bidirectional
 $\emptyset$: closed

Entrance-rate control $\mathbf{q}$
Piecewise admissible rates
 $q_C^+(t), q_C^-(t)$

Joint control variable
 $\mathbf{z} = (\mathbf{s}, \mathbf{q})$

3. Crowd Dynamics Simulator Bellman-Conservation Solver $\mathcal{S}(\mathbf{z}; \hat{\mathbf{p}}, \eta_0)$



(a) Multi-stage / multi-route density decomposition
 $\rho = \sum_o \sum_r \rho_{o,r}$



(b) Total density and speed update
Density-dependent speed
 $f(\rho)$ decreases with ρ



(c) Direction constraints and guidance
Allowed directions $\mathcal{U}(x; \mathbf{s})$
Anisotropic metric $M(x; \eta_0)$



(d) Constrained Bellman equation
Compute potential and optimal direction



(e) Finite-volume conservation update
Advance $\rho_{o,r}$ using upwind fluxes



(f) Internal entrance flux control
Actual flux capped by $q_C^\pm(t)$



(g) Stage transition update
Route switching governed by $\hat{\mathbf{p}}$

Simulation Outputs



Density field
 $\rho(x, t)$



Potential field
 $\phi(x, t)$



Velocity field
 $\mathbf{v}(x, t)$



Entrance wait blocking B



Channel throughput
 R_C

4. Performance Evaluation

- Efficiency J_1
travel time
- Safety J_2
density exposure
- Load balance
 $J_3 = \text{Var}(R_C)$
- Waiting J_4
- Smoothness J_5

Weighted objective

$$J = \sum_{k=1}^5 \lambda_k \tilde{J}_k$$

5. Upper-level HCMBO Optimizer

1 Feasible directions
 $\mathbf{s}_{\text{cand}}$

2 Capacity map
 $q = T_S(x)$

3 Search domain
 $\mathbf{x} \in [0, 1]^d$

4 Multi-fidelity simulation eval.

5 Final re-eval.
select $\mathbf{z}^*$

Best policy under limited simulation budget

feedback update $\mathbf{z} = (\mathbf{s}, \mathbf{q})$